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Wing loading and habitat selection in forest beetles: Are red-listed species poorer dispersers or more habitat-specific than common congeners?

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ABSTRACT

Traits such as poor dispersal ability and high habitat specificity are thought to predispose some species to a greater risk of extinction than others. Habitat preferences and morphological features associated with dispersal ability were compared between red-listed species and common congeners co-occurring in boreal forests in northern Sweden. Measurements of body size, wing loading and wing aspect ratio were used to compare dispersal abilities, while catches from experimental treatments of dead wood were used to compare habitat preferences. We also compared how restricted red-listed species were to particular sites or habitats relative to common species. The red-listed *Epuraea longipennis* was longer and wider, while *Epuraea deubeli* weighed less than common *Epuraea* species. In contrast to expectations, these red-listed species had a larger wing area (relative to their body mass) and thus a lower wing-loading than congeneric species, suggesting superior dispersal abilities. The red-listed *Tachinus elegans* possessed intermediate morphological characteristics, compared with common congeners. However, the relationships between the risk of extinction in fragmented habitats and size and dispersal ability are likely to be scale-dependent, so intermediate or superior dispersal abilities may increase extinction risk at some scales. Red-listed species were not found in fewer sites or habitat types than congeners so were not more likely to be habitat specialists. However, some red-listed species preferred deeply shaded and *Fomitopsis pinicola*-inoculated logs, relative to congeners, suggesting that specificity to these particular microhabitats may be connected with extinction risk.

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1. Introduction

Traits such as a low or fluctuating population size, rarity, poor dispersal ability and habitat specialisation may predispose some species to extinction (Henle et al., 2004) and species with combinations of extinction-promoting traits may be particularly vulnerable (Davies et al., 2004). Red-lists include species that are considered to be threatened because their populations are small or shrinking (Gärdenfors, 2001) and

the primary threat to populations of many red-listed species in Scandinavia is the habitat loss and reduction in habitat quality caused by forestry (Stokland, 1991; Berg et al., 1994; Thor, 1998).

Intensive forestry has changed the composition and connectivity of forest habitats in Sweden considerably and less than 1% of the productive forest land is currently protected (Linder and Östlund, 1992; Löfgren, 1997; Anon., 2001). While the landscape was historically dominated by older forests, it

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also contained a patchwork of successional stages recovering after fire and other natural disturbances (Östlund et al., 1997). In the modern landscape, even-aged stands in earlier successional stages cover the majority of the landscape (Linder and Östlund, 1998; Axelsson and Östlund, 2001), creating a matrix of habitat that may not be habitable for those species that are dependent on older stand types. In order to survive in the modern landscape in the long-term, species dependent on older stand types must disperse through the matrix and successfully colonise new patches (Harrison and Bruna, 1999), or survive by spreading between connected patches before they are harvested.

Successful colonisers are characterised by superior dispersal abilities and/or high reproductive rates, although there may be some trade-offs between these traits (Rankin et al., 1994; Langelotto et al., 2000). Species relying on more persistent habitats, such as hollow trees (Nilsson and Baranowski, 1997; Ranius and Hedin, 2001) or stable bogs (Wikars, 1997) are expected to possess inferior dispersal abilities to those relying on less persistent habitats, such as recently fallen coarse woody debris (CWD) (Nilssen, 1984; Solbreck, 1980) and burned areas (Wikars, 1997), because short patch persistence times select for better dispersal abilities (Southwood, 1977; Nilsson and Baranowski, 1997). However, long-lived habitats are those most likely to disappear from the managed forests of northern Sweden, where the rotation cycles are approximately 120 years (Ranius and Kindvall, 2004). Species that are poor dispersers and dependent on long-lived habitats are thus most likely to become extinct in the managed landscape.

For insects, particularly those that are rare, dispersal can be difficult to measure directly or in laboratory situations (although see Jonsson, 2003). Fortunately, flight morphology and flight performance are associated, so morphology provides a useful indicator of dispersal ability (Berwaerts et al., 2002). In particular, a low wing loading (body mass divided by wing area), is thought to represent a superior flying ability because flight becomes more energetically efficient with decreasing wing loading (Angelo and Slansky, 1984). Long, narrow wings, which have a high aspect ratio (wing length divided by wing width), adapt a species to fast flapping flight (Norberg, 1990), while broad wings adapt a species to gliding flight (Wootton, 1992) and mayflies with short wings have been shown to occupy fewer sites than those with large wings (Malmqvist, 2000). Large body size improves dispersal ability because it reduces the mass-specific cost of flight (Roff, 1991; Berwaerts et al., 2002). Additionally, large insects maintain a higher thoracic temperature than small insects and may thus have greater flight musculature efficiency (Chapman, 1998).

Of the approximately 1000 saproxylic beetles in Sweden, 474 are red-listed (Gärdenfors, 2000). Apart from information on distributions, obtained through surveys, very little information is available on the ecology of red-listed saproxylic beetles in Sweden. Both landscape-scale habitat loss and the loss of coarse woody debris habitats within remaining forests are considered to be major threats to these species (Ökland et al., 1996; Jonsell et al., 1998; Rukke, 2000). Red-listed species have either been observed to decline in abundance or have suffered destruction of their habitats and are thus

considered to be at risk of becoming extinct from a region (Gärdenfors, 1996). Red-listed species are thus useful for testing *a priori* hypotheses concerning those characteristics that are typical of rare and threatened species. In this study, we use morphological characteristics and experimental substrate manipulations to determine which traits, if any, are characteristic of red-listed species. In particular, we hypothesize that red-listed species in boreal forests are poorer dispersers and have more specialised habitat preferences than common congeners. A comparison of a red-listed species with several of its congeners should allow us to better understand the subtle differences that result in one species being threatened by forestry, while a set of related species remain unthreatened.

Extensive collection of saproxylic species from experimental logs over a period of 3 years has provided us with an unusually large sample size for a number of red-listed species in a controlled experiment. We thus have a unique opportunity to specifically test hypotheses concerning dispersal morphology and habitat use. We address the following questions:

1. Do red-listed beetles have morphological characteristics (body size, wing loading and aspect ratio) that make them less efficient dispersers than their more common relatives?
2. Do red-listed species occupy fewer sites and habitat types than common congeners?
3. Do red-listed species differ from common congeners in terms of their substrate and stand type preferences?

We tested questions one and three for three red-listed species and common congeners and question two for six red-listed species.

2. Methods

2.1. Study area

Beetles were collected from sites within the central-boreal vegetation zone of Sweden (Ahti et al., 1968), between 63.620° N and 64.285° N and 16.889° E and 20.132° E and ranging from 100 to 550 m above sea level. In total, 27 sites were sampled: nine in unmanaged Norway spruce (*Picea abies*) dominated forest stands in nature reserves and national parks, nine in nearby mature managed forest and nine in nearby clear-cut stands. The mature managed forests had probably been selectively logged for trees of large diameter, but never clear-cut. All “clear-cut” sites had been cut between 1999 and early 2001 (see Gibb et al., 2005 for further details). Old-growth forests (mean 72.5 m³ ha⁻¹) contained considerably more CWD than both mature managed forests (mean 24.3 m³ ha⁻¹) and clear-cuts (mean 13.6 m³ ha⁻¹), particularly with respect to older and larger diameter logs and snags (Gibb et al., 2005).

2.2. Trapping methods

Insects were collected from logs of Norway spruce and birch, *Betula pubescens* and *B. pendula*, which had been placed out between September 2001 and March 2002. Logs were 4 m in

length and approximately 20 cm in diameter and had been produced in the same felling operation. In each of the 27 sites, three blocks of spruce (each containing five logs and one high stump and its sawn-off top) and two blocks of birch (5 logs each) were sampled. The logs were placed under the following treatments: control; shaded; burned; and inoculated with fungus. Inoculations were performed using one cellulose-decomposing and one lignin-decomposing fungus species for both spruce (*Fomitopsis pinicola* and *Resinicium bicolor*) and birch (*Piptoporus betulinus* and *Fomes fomentarius*).

Trunk eclector traps (covering an area of approximately 0.2 m²) were used to collect beetles emerging from the wood, while window traps (perspex pane: 10 × 15 cm, attached directly to the logs) were used to collect beetles attracted to or flying close to the logs (see Johansson et al. (2006) for further details about the traps). We restricted ourselves to sampling from our experimental logs in order to maintain the experimental aspect of this study. The species on which we focussed were found in substantial numbers in the substrates we examined, so substrate preference in the context of coarse woody debris is relevant for these species, although it probably does not cover all possible habitats for all species. Trapping was conducted over five time periods: July–August 2002; September 2002–May 2003; May–June 2003; June–July 2003; and July–September 2003. Window traps were used in all periods, except September 2002–May 2003 and July–September 2003. Samples collected from one additional nearby locality (also including a nature reserve, mature managed forest and clear-cut stand) using similar methods in four trapping periods in 2001 were also used for the morphological comparisons.

All beetles were extracted from the samples and determined to species by experts (S. Lundberg and J. Hilszczański). The most common of the red-listed species and common congeners were selected for analysis.

2.3. Red-listed species

Red-listed species were selected using the Swedish Red List (S.L.U., 2005, <http://www.artdata.slu.se>). All red-listed species considered in this study are categorised as “near-threatened”, except for *Epuraea longipennis* (Sjöberg 1939), which is categorised as “data deficient” because it has previously been found in very few localities. The majority of these species are red-listed as a result of declines in abundance or destruction of their habitats, rather than intrinsic rarity. They are described below.

Agathidium mandibulare (Sturm 1807) (Leiodidae) develops mainly in slime moulds on and in lying dead wood of large diameter. *A. nigrinum* (Sturm 1807) lives in rotten, fungus-infested deciduous logs. Both species live in dense forests. They are red-listed (category A2c, IUCN, 2001) because they are thought to have decreased in abundance by >15% in the last 10 years due to habitat loss (S.L.U., 2005).

Atomaria alpina (Heer 1841) (Staphylinidae) occurs mainly in old spruce stands with a high abundance of the red-banded polypore (*F. pinicola*). It occurs over an area of less than 2000 km² and, as a result of habitat loss and fragmentation, the species has decreased by >15% in abundance in the last 10 years and is red-listed under the categories A2c, B2ab(iii).

Epuraea deubeli (Reitter 1898) (Nitidulidae) is thought to live in the tracks of the bark beetles *Trypodendron lineatum* (Olivier 1795) and *Ips typographus* (Linnaeus 1958). It is believed to live on either the ambrosia fungus farmed by *T. lineatum* or the larvae of *T. lineatum* (Gyllenhal 1827) (S.L.U., 2005). The species is red-listed because it has a small area of occupancy (known from just 11 locations), with declining habitat quality (category B2b(iii), IUCN, 2001). Removal of CWD for protection against pests before the CWD can become suitable habitat is considered the primary threat to this forest-living species (S.L.U., 2005).

Epuraea longipennis had previously been found only in a total of three locations in the Jämtland and Ångermanland provinces. Records from this study indicate that it is also found in the Västerbotten and Lappland provinces (R. Pettersson, unpublished). It is usually collected from bark beetle-attacked spruce in old forests (S.L.U., 2005).

Tachinus elegans (Eppelsheim 1893) (Staphylinidae) is thought to be associated with old, moist spruce stands. This species is red-listed because it has a small area of occupancy (known from just four locations), with declining habitat quality (category B2b(iii), IUCN, 2001). It can occur in rotting mushrooms, the felled tops of spruce trees, in moss and in birch stumps and is thought to be threatened by the loss of humid, natural spruce forests (S.L.U., 2005).

2.4. Morphometric measurements

Morphometric measures thought to be related to dispersal ability were compared between red-listed species and three co-occurring common congeneric species. The red-listed *E. longipennis* and *E. deubeli* were compared with *E. angustula* (Sturm 1844), *Epuraea boreella* (Zetterstedt 1828) and *Epuraea pygmaea* (Gyllenhal 1828) and the red-listed *T. elegans* was compared with *T. laticollis* (Gravenhorst 1802), *T. pallipes* (Gravenhorst 1806) and *T. proximus* (Kraatz 1855). The *Epuraea* specimens were collected in eclector traps, while the *Tachinus* specimens were collected in window traps. The only exception to this was *E. deubeli*, which was collected almost exclusively in window traps. In order to make comparisons between *E. deubeli* and the other species possible, 20 specimens of *E. pygmaea* from the window traps were also measured, in order to determine the difference in weight resulting from the individuals caught in window traps having expended considerable amounts of energy flying. *Epuraea pygmaea* from the eclector traps were of similar lengths, but weighed approximately 1.11 times as much as *E. pygmaea* specimens from the window traps. This number was used as a correction factor for the weight of the *E. deubeli* specimens collected from window traps so that they could be compared with *Epuraea* specimens collected in eclector traps.

Body length and width, wing length and width, wing area and body mass were measured for 20 males of each species. Body length and width were measured to the nearest 0.1 mm at the longest and widest points of the insects using a measuring slide (Carl Zeiss, Germany) under a dissecting microscope. The right wing was removed and placed on a slide under a cover slip. *Epuraea* specimens were then dried for 8 h and *Tachinus* specimens (which were larger) were dried for 16 h at 70 °C in a drying oven. Samples were weighed to

the nearest 0.001 mg on electronic laboratory scales (Mettler Toledo ME30, Stockholm).

Wings were placed under a light microscope connected to a computer screen and lit from below, such that their outlines were clearly visible. Clear images of each of the wings were captured and saved using the program Kappa Image Base Control 2.7.2 (Kappa opto-electronics GmbH, Gleichen, Germany, 2003). The program Image Tool for Windows (Version 3.0, The University of Texas Health Science Centre in San Antonio) was used to measure the length, width and area of the wings.

Aspect ratios for wings are calculated by dividing wing length by wing width, while wing loading is calculated by dividing the dry mass by the total wing area (Vogel, 1981).

2.5. Data analysis

2.5.1. Do morphometric measures, wing loadings or aspect ratios differ between red-listed species and common congenics?

Differences between the six morphometric measures: body length and width, wing length, width and area and body mass of species within the genera *Epuraea* and *Tachinus* were tested separately for each genus using MANOVA, followed by ANOVAs on subgroups, when the initial MANOVA was significant. The Pillai's Trace test statistic was used to test the null hypothesis that there were no differences in species group centroids because it is the most robust of the available tests (Johnson and Field, 1993). Significant ANOVAs were followed by Tukey's HSD tests, which compared amongst species. Data that did not meet the assumptions of MANOVA were $\log(x + 1)$ transformed such that they did.

Species were compared in terms of their aspect ratios using ANOVA and wing areas using ANCOVA. Body mass was used as a covariate because wing loading is expected to increase with size across species (Dudley, 2000). All analyses were performed on JMP 5.1 (SAS, 2003).

2.5.2. Do red-listed species occupy fewer sites and habitats than common congenics?

The six most-common red-listed species in our data set were selected (those of which we collected more than 20 individuals over the entire trapping period 2002–2003: *A. mandibulare*, *A. nigrinum*, *A. alpina*, *E. deubeli*, *E. longipennis* and *T. elegans*) and their distributions amongst sites and habitats and substrates were compared with those of the distributions of common congenics. Between three and six congeneric species were more common than each red-listed species and could thus be used to estimate expected site- or habitat-specificity in the analysis. For the analysis of site-specificity, we counted the three stand types within each area as separate sites, such that 27 sites were used in total. For the combined analysis of habitat- and substrate-specificity, the seven spruce and five birch treatments and the three stand types were included, such that a maximum of 36 possible habitat–substrate combinations could be occupied.

Red-listed species were collected in fewer traps than common congeneric species so we randomly selected a specific number of traps from the total in which each common species was collected. The number of traps was thus equal for

each species in each red-listed–non-red-listed comparison, but differed between comparisons, depending on the abundance of the particular red-listed species. This correction allowed us to rule out the possibility that red-listed species could be found to be more site- or habitat-specific, simply as a result of random sampling. The hypotheses that red-listed species were found: (a) at fewer sites and (b) in fewer stand types and substrates than non-red-listed species were tested using a Chi-square test. Expected frequencies were the average of the frequencies for related non-red-listed species.

2.5.3. Do the substrate and stand type preferences of red-listed species differ from those of common congenics?

The relative abundances of the red-listed *Epuraea* and *Tachinus* species were compared with those of common congenics using ANOVA with the factors “stand type” (fixed), “log treatment” (fixed) and “species” (random). Data were standardised (such that the sum of each column = 1) so that differences in the overall abundances of each species did not contribute to differences between log treatment and stand types. This proportional data was arcsine-transformed. The interaction terms “species * stand type”, “species * log treatment” and “species * stand type * log treatment” were of interest because these terms indicate whether different species respond differently to different stand types and log treatments. “Species” was a random factor and F-values and p-values were corrected by using the appropriate MS denominators for a model with two fixed and one random factor (see Quinn and Keough, 2002, p. 257). Type III sums of squares were used so that our results were not dependent on the order of the terms in the model. *Epuraea* species were much more prevalent on spruce than birch logs, so patterns were tested separately for eclector- and window-trapped specimens collected from spruce. *Tachinus* species were collected predominately in window traps, but were common on both tree species, so data were analysed separately for window-trapped specimens on birch and spruce logs.

3. Results

3.1. Do morphometric measures, wing loadings or aspect ratios differ between red-listed species and common congenics?

The measured morphometric characteristics (body length and width, wing length, width and area and body mass) differed between species for both *Epuraea* (MANOVA: $F_{(20,376)} = 12.97$, $p < 0.001$) and *Tachinus* (MANOVA: $F_{(15,222)} = 19.10$, $p < 0.001$). ANOVAs (Tables 1 and 2) showed that significant differences occurred among species for all morphometric characteristics. Patterns for red-listed species were inconsistent: *E. deubeli* was smaller than all other *Epuraea* species except *E. pygmaea* for most measures and also weighed the least, while *E. longipennis* was the largest for all measures, although its weight was quite similar to most other *Epuraea* species (Table 2). *T. elegans* proved intermediate for all measures (Table 2).

Aspect ratios were similar for red-listed and common species, with no apparent patterns (Tables 1 and 2, Fig. 1). When the effects of body mass on wing area were included in an

Table 1 – ANOVA testing each of the morphometric characteristics of *Epuraea* and *Tachinus* from the MANOVA (body length and width, wing length and width and body mass) and aspect ratio

Source	<i>Epuraea</i> (df = 4,95)			<i>Tachinus</i> (df = 3,76)		
	Mean square	F	p	Mean square	F	p
Body length^a						
Species	0.2935	36.28	<0.0001	50.2208	188.83	<0.0001
Error	0.0081			0.2660		
Body width						
Species	0.3637	15.41	<0.0001	2.1545	266.03	<0.0001
Error	0.0236			0.0081		
Wing length^b						
Species	2.0506	47.16	<0.0001	0.7756	409.80	<0.0001
Error	0.0435			0.0019		
Wing width^b						
Species	0.1866	20.47	<0.0001	0.6252	311.72	<0.0001
Error	0.0091			0.0020		
Body mass^b						
Species	0.0906	20.49	<0.0001	7.4176	57.27	<0.0001
Error	0.0044			0.1295		
Aspect ratio						
Species	0.0831	5.75	<0.0003	0.1142	42.13	<0.0001
Error	0.0144			0.0027		

Results of post hoc Tukey's tests are shown in Table 2.

a,b Data $\log_{10}(x + 1)$ transformed for *Epuraea* and *Tachinus* species, respectively.

Table 2 – Mean $\pm$ SE for the morphometric measurements of *Epuraea* and *Tachinus* species

Species	Body		Wing		Body mass (mg)
	Length (mm)	Width (mm)	Length (mm)	Width (mm)	
<i>E. angustula</i>	2.87 $\pm$ 0.05 cd	1.10 $\pm$ 0.02 bc	3.36 $\pm$ 0.04 b	1.29 $\pm$ 0.02 b	0.333 $\pm$ 0.014 a
<i>E. boreella</i>	3.31 $\pm$ 0.08 b	1.22 $\pm$ 0.03 ab	3.54 $\pm$ 0.05 b	1.34 $\pm$ 0.02 ab	0.336 $\pm$ 0.013 a
<i>E. deubeli</i>	3.06 $\pm$ 0.06 bc	1.00 $\pm$ 0.03 c	3.06 $\pm$ 0.06 c	1.17 $\pm$ 0.03 c	0.204 $\pm$ 0.016 b
<i>E. longipennis</i>	3.65 $\pm$ 0.08 a	1.35 $\pm$ 0.06 a	3.75 $\pm$ 0.05 a	1.39 $\pm$ 0.03 a	0.322 $\pm$ 0.018 a
<i>E. pygmaea</i>	2.66 $\pm$ 0.04 d	1.23 $\pm$ 0.02 ab	2.99 $\pm$ 0.03 c	1.18 $\pm$ 0.02 c	0.340 $\pm$ 0.015 a
<i>T. elegans</i>	7.84 $\pm$ 0.13 ab	2.03 $\pm$ 0.02 a	5.85 $\pm$ 0.05 b	2.44 $\pm$ 0.02 b	1.27 $\pm$ 0.09 b
<i>T. laticollis</i>	4.75 $\pm$ 0.11 c	1.36 $\pm$ 0.02 b	4.14 $\pm$ 0.04 c	1.76 $\pm$ 0.02 c	0.41 $\pm$ 0.03 c
<i>T. pallipes</i>	7.68 $\pm$ 0.09 b	1.98 $\pm$ 0.02 a	5.83 $\pm$ 0.05 b	2.45 $\pm$ 0.03 b	0.98 $\pm$ 0.10 b
<i>T. proximus</i>	8.17 $\pm$ 0.12 a	2.02 $\pm$ 0.02 a	6.53 $\pm$ 0.08 a	2.58 $\pm$ 0.03 a	1.58 $\pm$ 0.08 a

Results of post hoc Tukey's tests for each of the univariate ANOVAs are given: means with the same letter are not significantly different from one-another. Red-listed species are shown in bold-face type.

ANCOVA, wing area proved to be greatest for the two red-listed *Epuraea* species (although not different from *E. boreella*) and intermediate for the red-listed *T. elegans* (Table 3, Fig. 2). *E. longipennis* was the only red-listed species that had a greater wing area (and thus lower wing loading) than expected for its body mass (Fig. 2).

3.2. Do red-listed species occupy fewer sites or habitats than common congeners?

Red-listed species were not more restricted in their distributions amongst sites than related non-red-listed species ($\chi^2_{(5)} = 1.42$, $p = 0.922$), occupying between 10 and 24 of the 27 sites. Nor were red-listed species more restricted in their distributions amongst forest and substrate types than related

non-red-listed species ($\chi^2_{(5)} = 3.91$, $p = 0.562$), occupying between 10 and 23 of the 36 possible habitat-substrate types.

3.3. Do the substrate and stand type preferences of red-listed species differ from those of common congeners?

ANOVA revealed significant interactions between species and treatment for *Epuraea* species captured in both eclector and window traps on spruce (Table 4, Fig. 3). For the eclector-trapped *Epuraea* (*E. deubeli* was too rare in eclector traps to be included in these analyses), simple effects tests showed that differences were significant for *Epuraea* captured on *F. pinicola* logs and snags. *F. pinicola* logs supported a significantly higher proportion, while snags supported a significantly lower proportion of *E. longipennis*, relative to common

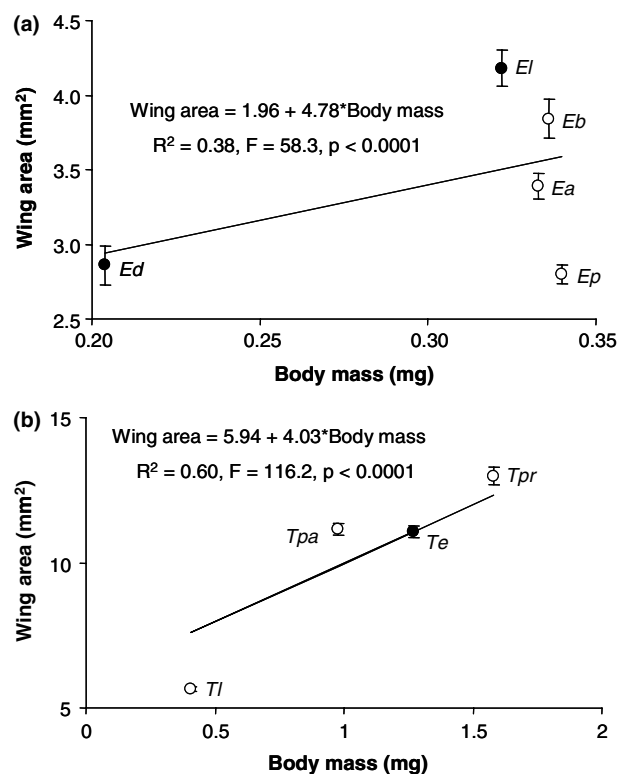
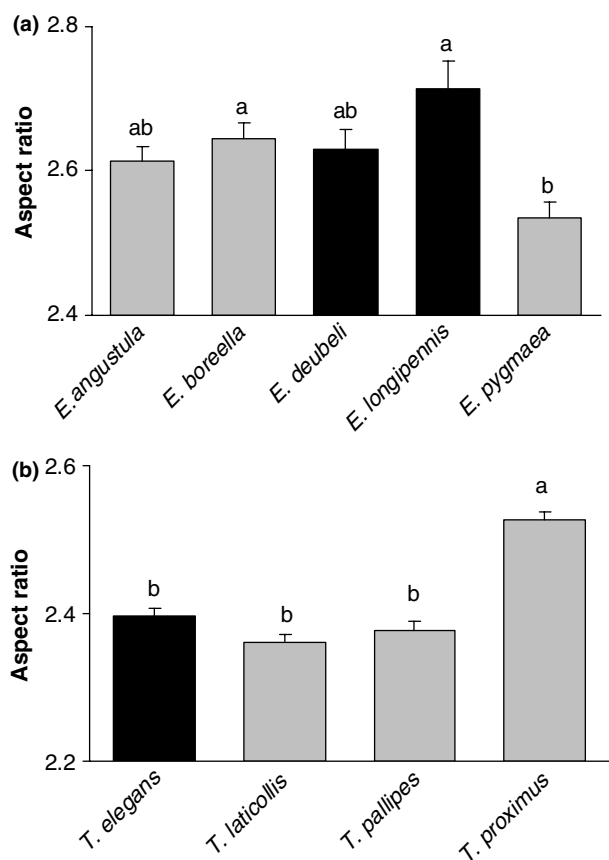


Fig. 1 – Mean $\pm$ SE aspect ratio for: (a) *Epuraea*, and (b) *Tachinus*. Red-listed species are shown with black bars.

congenerics. For the window-trapped *Epuraea*, simple effects tests showed that differences were significant only for the shade and snag treatments. On shaded logs, significantly more *E. longipennis* were captured than *E. angustula*, *E. deubeli* and *E. pygmaea*, while significantly fewer *E. longipennis* were captured on the snags.

For the *Tachinus* species, no significant interactions including species occurred for individuals collected in the window traps on birch (Table 4, Fig. 4). However, the three-way interaction between species, treatment and stand type was significant for *Tachinus* captured from spruce. For the old managed forest sites, *T. elegans* was proportionally less common on

Fig. 2 – Mean $\pm$ SE wing area for: (a) *Epuraea* ($Ea = E.$ *angustula*; $Eb = E. *boreella*; $Ed = E. *deubeli*; $El = E. *longipennis*; $Ep = E. *pygmaea*), and (b) *Tachinus* ($Te = T. *elegans*; $Tl = T. *laticollis*; $Tpa = T. *palliatu*s; $Tpr = T. *proximus*). Regression lines are for all individuals in all species. Symbols above the line have a lower wing loading than expected, based on their weight, while symbols below the line have a greater wing loading than expected. Black symbols represent red-listed species.$$$$$$$$

control logs and more common on shaded logs than common congeners.

4. Discussion

4.1. Functional morphology and dispersal

Amongst the red-listed species, wing loading ranged from medium to low, relative to common species. The red-listed

Table 3 – ANCOVAs testing for differences in wing area between species for *Epuraea* and *Tachinus*, with body mass as a covariate

Source	<i>Epuraea</i>				<i>Tachinus</i> ^a			
	df	SS	F	p	df	SS	F	p
Species	4	2.010	93.09	<0.0001	3	3.340	174.58	<0.0001
Body mass	1	1.039	192.28	<0.0001	1	0.046	7.22	0.0089
Error	94	0.507			75	0.478		
<i>E. longipennis</i> > <i>E. deubeli</i> = <i>E. boreella</i> > <i>E. angustula</i> > <i>E. pygmaea</i>					<i>T. proximus</i> > <i>T. pallipes</i> = <i>T. elegans</i> > <i>T. laticollis</i>			

Results from post hoc Tukey's tests are shown, with red-listed species shown in bold-face type.

a Wing area was log(x + 1) transformed.

Table 4 – Results of ANOVA comparing the standardised abundances of each species amongst log treatments and stand types

Source	df	Mean square	F	p
<i>Epuraea</i> , spruce, eclector traps				
Stand type	2	0.0022	11.12	0.010
Treatment	5	0.0010	3.29	0.033
Species	3	0.0000	<0.01	1.000
Stand type * treatment	10	0.0002	1.79	0.105
Stand type * species	6	0.0002	1.17	0.322
Treatment * species	15	0.0003	1.84	0.027
Stand type * treatment * species	30	0.0001	0.84	0.719
Error	576	0.0002		
<i>Epuraea</i> , spruce, window traps				
Stand type	2	0.0032	12.57	0.003
Treatment	5	0.0034	3.62	0.017
Species	4	0.0000	<0.01	1.000
Stand type * treatment	10	0.0005	1.72	0.110
Stand type * species	8	0.0003	0.84	0.572
Treatment * species	20	0.0009	3.13	<0.001
Stand type * treatment * species	40	0.0003	1.00	0.478
Error	720	0.0003		
<i>Tachinus</i> , birch, window traps				
Stand type	2	0.0075	14.07	0.002
Treatment	4	0.0025	4.00	0.020
Species	4	0.0000	<0.01	1.000
Stand type * treatment	8	0.0028	4.68	0.001
Stand type * species	8	0.0005	0.65	0.738
Treatment * species	16	0.0006	0.77	0.718
Stand type * treatment * species	32	0.0006	0.73	0.866
Error	600	0.0008		
<i>Tachinus</i> , spruce, window traps				
Stand type	2	0.0074	38.17	<0.001
Treatment	5	0.0018	2.80	0.045
Species	4	0.0000	<0.01	1.000
Stand type * treatment	10	0.0008	1.41	0.213
Stand type * species	8	0.0002	0.47	0.879
Treatment * species	20	0.0007	1.57	0.054
Stand type * treatment * species	40	0.0006	1.43	0.044
Error	720	0.0004		

When interactions were significant, simple effects tests were used to test the effect of species at each level of treatment or stand type. Post hoc t-tests (reported in the text) were used to test the a priori hypothesis that habitat preferences differed between species. Values of $p < 0.05$ are considered significant.

Epuraea, *E. longipennis* and *E. deubeli*, had relatively large wing areas (and thus low wing loadings) compared with common congenics, after body mass was accounted for. A low wing-loading is associated with a superior flying ability (Wainright, 1994; Berwaerts et al., 2002), suggesting that both species may be good dispersers, relying on resources with short persistence times (McLachlan, 1985; Wikars, 1997; Denno et al., 2001), such as the young CWD on which they were collected, or highly scattered resources. The relatively high aspect ratios of these species also suggest that their wing shape predisposes them to flying longer distances than other *Epuraea*.

While the relationship between dispersal ability and the risk of extinction as a result of habitat fragmentation may appear monotonic over some scales of observation, previous studies have noted that the fragmentation of breeding habitat may cause maximum effects at intermediate dispersal power, rather than at low dispersal power, suggesting that the rela-

tionship is unimodal, rather than monotonic (Fahrig, 1998; Thomas, 2000). This is because poorer dispersers respond to their habitat as though it is fragmented, even before anthropogenic fragmentation occurs, while better dispersers may never perceive the habitat as being fragmented. It is thus possible that species with superior dispersal morphologies, such as *E. deubeli* and *E. longipennis*, or those with intermediate dispersal morphologies, such as *T. elegans*, are affected more by habitat fragmentation than similar species with poorer dispersal abilities, depending on where they lie on this unimodal curve.

Links between size and vulnerability to extinction have previously been drawn, however it is unclear whether large or small species are likely to be at a greater risk of extinction (Pimm, 1991; Lawton, 1994; Gaston and Blackburn, 1995). *E. longipennis* was wider and longer, while *E. deubeli* was lighter than common congenics. Species of intermediate size may be at least risk of extinction because a positive

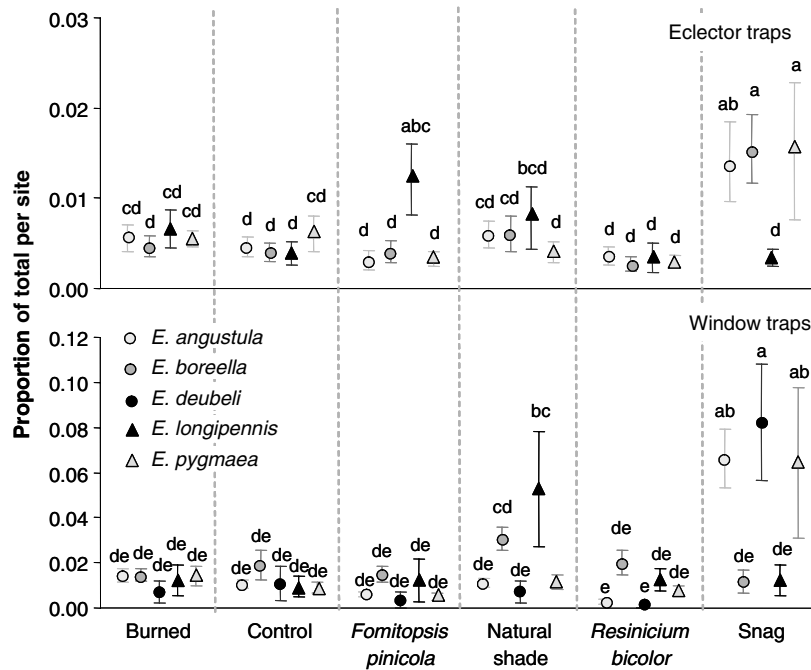


Fig. 3 – Mean $\pm$ SE standardised abundance per site of the common and red-listed *Epuraea* species in eclector and window traps on spruce. Symbols with the same letters are not significantly different. Black symbols represent red-listed species.

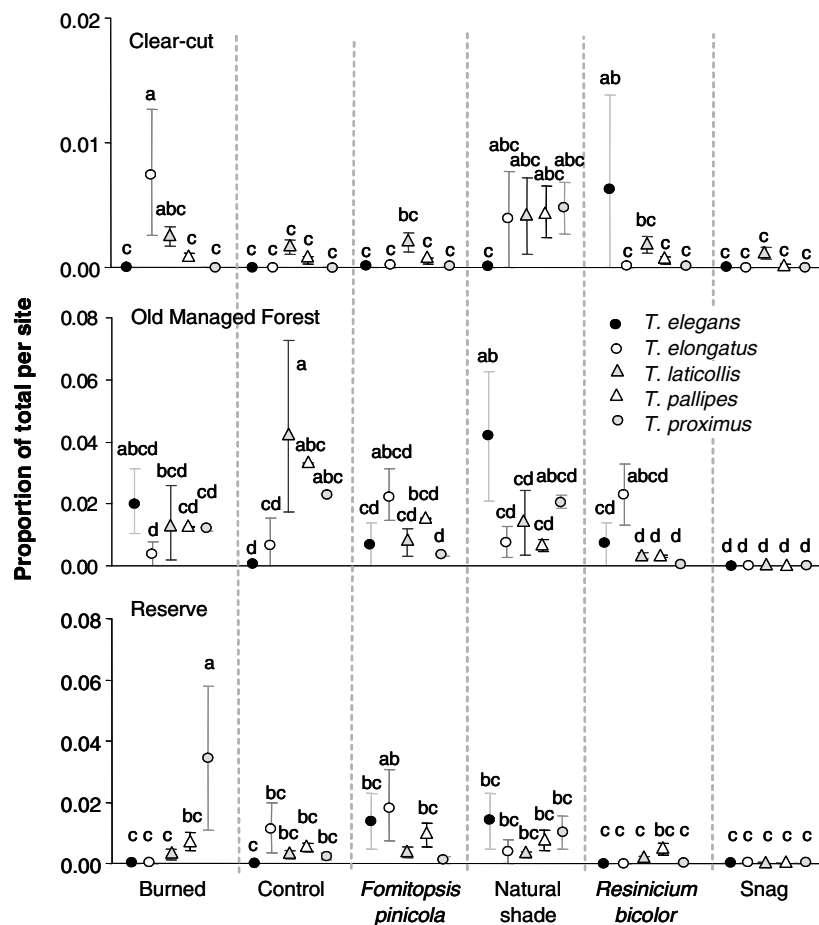


Fig. 4 – Mean $\pm$ SE standardised abundance per site of the common and red-listed *Tachinus* species collected in window traps on spruce in clear-cuts, managed forests and reserves. Symbols with the same letters are not significantly different. Black symbols represent red-listed species.

correlation between body size and extinction risk, due to population size and rate of recovery (Lawton, 1994), is counteracted by a negative correlation between body size and extinction risk, due to fluctuations in population size (Pimm, 1991). This argument could explain why the two red-listed *Epuraea* species are unusually large or small. However, *T. elegans* was of intermediate dimensions, so size is an unlikely explanation for the threatened status of this species.

Previous studies have failed to find a link between the body size of beetles and their response to habitat fragmentation (Didham et al., 1998; Davies et al., 2000; Driscoll and Weir, 2005) and the relative importance of forces increasing the risk for small or large species may be scale-dependent (Blackburn and Gaston, 1997). Further study is required to determine the circumstances under which species of greater or lesser or even intermediate size or dispersal ability might be most vulnerable. Importantly, the sizes and flying abilities examined in this study may not have covered a sufficient range of values to observe consistent patterns because small insects may rely more on wind for dispersal than large insects and there can be considerable scatter in scaling relationships for insect flight (Ellington, 1999). Further studies may need to go beyond our within-genus approach, or to focus on genera with a greater range of body sizes or on other dispersal-related characteristics, such as flight behaviour or wing musculature (e.g. Jonsson, 2003).

4.2. Site- or habitat-specificity

In contrast to expectations, red-listed species were not more restricted in their distributions amongst sites or habitats than common congeners, suggesting that they are not more selective of substrates or habitats and do not have more limited distributions in the study area. The red-listed species selected for this study were those that were the most common in the traps, so it is likely that they belong to a group of species that are locally abundant, but with a restricted range, rather than being widespread but locally rare, and this may explain why they were no more habitat-specific than common congeners. Additionally, the experimental CWD was in the short-lived early decay stage and species that depend on such ephemeral resources are known to be good dispersers (Nilssen, 1984; Forsse and Solbreck, 1985), so they may suffer less from fragmentation of their habitats than species that are dependent on the intermediate or later stages of decay.

Modern forestry reduces the diversity of CWD objects in managed forests relative to that in unmanaged forests (Rouvinen et al., 2002) and previous studies show that saproxylic beetles can be highly specific to standing or lying wood and wood of particular sizes, decay stages and species (Berg et al., 1994; Jonsell et al., 1999; Jonsell and Weslien, 2003; Siitonen and Saaristo, 2000; Gibb et al., 2006). Additionally, many saproxylic beetles are dependent on particular species of wood-decaying fungi (Jonsell and Nordlander, 2002) or on burnt CWD (Lundberg, 1984; Wikars, 1997). Habitat specialists are expected to have a less continuous distribution than generalists and thus be at greater risk of extinction due to random sampling of the remaining habitat occurring during fragmentation or consequent habitat degradation (Sarre et al., 1996; Andr  n et al., 1997). While red-listed species were

not more restricted in their habitat use than is typical for their genus, it is possible that the substrates they prefer are those that are most threatened by forestry.

4.3. Utilisation of substrates and stand types

Of the CWD treatments, only shaded logs (*T. elegans* and *E. longipennis*) and logs inoculated with *F. pinicola* (*E. longipennis*) favoured the most common red-listed species relative to common congeners. *E. longipennis* was also relatively rare on snags. The composition of beetle assemblages on standing dead wood is strongly influenced by the presence of fungi (e.g. *F. pinicola*, Jonsell et al., 2005) and all *Epuraea* species examined in this study are thought to be facultative fungivores, living mainly in cambium microhabitats (R. Pettersson, unpublished data). A previous study (Jonsson, 2003) suggested that the red-listed *Oplocephala haemorrhoidalis* fared worse than its common relative *Bolitophagus reticulatus* in exploiting a scattered fungus resource, *F. fomentarius*, in forestry-affected areas as a result of poor dispersal and colonisation abilities. In contrast, evidence from our study of the morphology of *E. longipennis* suggests that this red-listed species is well-adapted for movement amongst scattered resources, relative to common congeners.

All *Epuraea* and *Tachinus* species examined in this study preferred old managed forests and reserves over clear-cuts (unpublished data). For *Tachinus* spp., suitable moist fungal and detrital microhabitats are probably more numerous in older forests than on clear-cuts. Old selectively logged managed forests supported similar numbers of most species to reserves, suggesting that they may provide habitat of sufficient quality and temporal continuity to support intact assemblages of saproxylic beetles over long periods of time, despite relatively low volumes of CWD (Gibb et al., 2005). However, the substrate preference of the red-listed *T. elegans* differed from common congeners in the old managed forest, where it preferred the shaded logs, suggesting that this species is affected by structural differences between old-growth and managed forests, e.g. in terms of age structure, structural complexity and tree density (  stlund et al., 1997;   kland et al., 2003), that affect exposure to sunlight. Our findings show that this species is more selective in its habitat use in managed than old-growth forests, and this greater selectivity may limit its ability to use mature managed forests.

5. Conclusions

Our findings are in concurrence with previous studies (Henle et al., 2004; Davies et al., 2004), in that they demonstrate that knowledge of single ecological (or morphological) traits is not sufficient to predict the sensitivity of all species to fragmentation or other human impacts. While red-listed species were not generally more selective in the number of habitats they occupied, a preference for habitats that may be threatened by forest management was shared by two of the three species tested. *T. elegans* was more common in the deep shade provided in reserves and shaded logs in managed forests, and *E. longipennis* was associated with the red-banded polypore, *F. pinicola*, but is probably not as spatially restricted as previously thought.

Both red-listed *Epuraea* species also differed from common relatives in terms of dispersal ability (as assessed by body size, wing loading and wing aspect ratio), although not always in accordance with predictions. The lack of concurrence with predictions may be a result of the small range of body sizes over which this study was performed in our attempts to use congeneric species. Additionally, the relationships between size and dispersal ability and vulnerability to extinction may depend on the scale at which fragmentation occurs, so predictions for a particular species may be better made with substantial prior knowledge of the scale at which it uses its habitat.

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